

Research Statement

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Humans begin learning language from the moment they are born by observing the sounds and actions around them. Even though young children cannot explicitly explain the grammar of the language they use, they learn the relationships between recurring expressions and situations, allowing them to understand and produce appropriate sentences. Understanding language goes beyond knowing the meanings of individual words. The same expression may require entirely different actions depending on who says it, when it is said, and with what intention. Language is ultimately a means of guiding action, and acting appropriately requires the ability to understand the context in which an utterance is situated.

This characteristic is not limited to human language learning. Companion animals also learn associations between particular linguistic signals and actions or rewards. For example, a dog trained in an English-speaking environment may respond to the command “Sit” but not to the Korean expression “Anja,” even though both request the same action. This illustrates that the meaning of a word is not given independently, but is formed through repeatedly experienced environments and interactions. However, recognizing a linguistic expression and understanding what that expression means in a given context are different problems.

I experienced university life across the period before and after the emergence of ChatGPT, which allowed me to directly observe this distinction. When using early language models, the greatest difficulty was that they could generate grammatically natural sentences while misunderstanding what the user had actually requested and producing irrelevant answers. At the time, I regarded this merely as an inconvenience caused by insufficient model performance. However, as language models become connected to search engines, external documents, personal information, and tools, and evolve into agents capable of carrying out actions autonomously, misunderstanding context no longer remains a minor response error. If a model cannot distinguish between a user’s instruction and a sentence contained in external data, or if it deviates from the original objective as an interaction progresses, it may perform unintended actions and create security risks.

These concerns led me to the following questions: How do AI models represent not only the surface form of an input, but also the intentions and roles embedded within it, the provenance and trustworthiness of information, and the context that accumulates over time? Can we detect and control the moment when a model begins to depart from the user’s original intention by examining its internal representations?

1 Research Conducted During My Master’s Program

1.1 TST-Mamba

NLOS Transient Data as a Physical Language and Mamba-Based Selective Signal Learning

When I began conducting research in a computer vision laboratory, I came to view NLOS transient data as another form of language that a model must interpret. In non-line-of-sight (NLOS) imaging, a camera cannot directly observe the target scene. Instead, the hidden scene must be inferred from the arrival times and intensities of light that has undergone multiple reflections between a visible wall and a concealed object. Each measurement is highly noisy and ambiguous in isolation, yet the relationships among signals distributed across time and space indirectly encode the structure of the hidden scene. Just as a word or sentence in natural language is interpreted within the context of a conversation, a transient signal acquires physical meaning only through its surrounding spatiotemporal relationships.

I did not regard this task as a simple denoising problem. The central challenge was to enable the model to distinguish signals required for reconstruction from irrelevant disturbances while preserving the fine spatiotemporal structures needed to infer the hidden scene. To address this challenge, I focused on Mamba, a state-space model capable of efficiently learning long-range sequential dependencies.

Mamba processes inputs sequentially, accumulates prior information in a hidden state, retains information that is relevant to the current input, and suppresses information that is unnecessary. I interpreted this mechanism not as the independent processing of individual measurements, but as a process of forming context in which the meaning of the current signal is determined in relation to preceding information. In NLOS transient data, a single time bin does not provide sufficient information about the hidden scene. Determining whether a particular measurement is a valid photon response returned from an object or merely background noise requires jointly considering temporal changes before and after that measurement and signals from neighboring spatial locations. I therefore used Mamba to scan the temporal sequence at each spatial location and treated the final hidden state, in which information from the full temporal interval had been selectively accumulated, as a compressed bottleneck representation. This representation is not a simple average or pooling result, but a learnable summary formed by retaining or suppressing different information depending on the input. Spatial Mamba then sequentially processes the compressed bottleneck tokens so that the model can learn, in latent space, how the signal at each location relates to signals at neighboring locations. In this architecture, Temporal Mamba forms context accumulated over time, whereas Spatial Mamba integrates context across spatial locations.

However, compressing context alone was not sufficient to faithfully reconstruct the original signal. The compressed representation provided the overall spatiotemporal structure but could not preserve every sharp temporal peak and weak photon return required for NLOS reconstruction. I therefore designed a structure that transfers the full temporal features through a skip path and combines them with the bottleneck code containing spatial context. Whereas the bottleneck path provides the global context needed to interpret the signal, the skip path preserves local and fine-grained evidence that may be lost during compression. By combining these two paths, the model can reconstruct a clean transient using both global relationships and detailed observational evidence, without relying solely on context or responding only to isolated signals. Through this analysis, I formalized TST-Mamba as a Mamba-based skip-connected denoising autoencoder. Temporal Mamba encodes a noisy transient sequence into a compressed representation, Spatial Mamba forms spatial context in latent space, and the skip-latent fusion decoder combines accumulated context with the original temporal details to reconstruct the signal. This work showed that Mamba can be interpreted not merely as an architecture for efficiently processing long sequences, but as a structure that learns what to remember, what to suppress, and which information to use for its final judgment.

Through this research, I came to believe that “context” does not simply mean providing a model with a large amount of information. Understanding context is a process of selecting the information required for the current task, forming relationships among information that is separated in time or space, suppressing irrelevant disturbances, and preserving the detailed evidence necessary for the final judgment. In NLOS transients, an incorrect selection may prevent the model from distinguishing signal from noise or, conversely, cause it to remove weak but valid signals. Similarly, in natural language models, misinterpreting the role or source of information may blur the distinction between user instructions and external data and cause the model’s behavior to deviate from its intended task. This experience expanded my interest beyond the restoration of physical signals toward the analysis of how context is formed and transformed within AI models and how those changes influence model decisions and behavior.

1.2 RiemannMamba

A Geometric Interpretation of Mamba’s Internal Representations

Through the TST-Mamba project, I became interested in how Mamba selectively retains and suppresses input information while forming task-relevant context in its hidden state. This led me to ask a further question: As Mamba processes an input sequence, which tokens does it treat as more important, and which tokens leave information in the internal state for longer periods so that it continues to influence subsequent decisions? In other words, even if all tokens are provided in the same order, might each token possess a different “duration” and degree of influence within the model?

The final output alone cannot reveal which tokens were quickly forgotten and which continued to affect later hidden-state changes. Existing explainable AI methods primarily estimate the importance of individual input features to the final output. In a state-space model such as Mamba, however, it is also necessary to understand when information is accepted, how long it is retained, and in what direction it changes as it is combined with

other information. I therefore viewed this process not as a collection of independent points in a high-dimensional representation space, but as geometric trajectories that progress at different rates for different tokens.

This perspective was inspired by the concept of proper time in Einstein’s general theory of relativity. In general relativity, time is not an absolute measure shared identically by every observer. The proper time experienced by an observer differs according to the geometry of spacetime shaped by mass and energy and the path followed by that observer. Encountering this concept led me to ask whether the tokens processed by Mamba might also possess an “internal time” distinct from their externally assigned order. All tokens pass through the same network depth and scan order, but the information contained in each token and its surrounding context differ. Some tokens may therefore induce only a small change in the hidden state and be processed with limited internal evolution, whereas ambiguous or complex tokens may produce larger changes in the model’s internal representations and continue to influence subsequent processing. Thus, even when the external computational steps are identical, the magnitude of change and the accumulated path experienced by the model in representation space may differ across tokens. I interpreted this as an internal time that flows at a different rate for each token.

This idea connects naturally to Mamba’s core variable, the input-dependent step size Δ . In Mamba, Δ adjusts the degree of state transition according to each input and determines how strongly the hidden state changes for each token. It has generally been interpreted as a computational gate that selectively passes or suppresses information. I instead considered whether Δ could serve as a geometric scale that determines how each change should be measured in the model’s hidden-state space. To formalize this idea, I treated the hidden-state space of each token as a Riemannian manifold and constructed state-dependent metric weights from Δ . The hidden-state change before and after the selective scan is then measured not as a simple Euclidean distance, but as a local line element weighted by this metric. By accumulating the changes produced at each layer, I defined the length of the path traveled by the model in representation space while processing an input as its internal time τ . Thus, τ does not represent a token index or actual computation time; it is the accumulated amount of geometry-weighted hidden-state evolution induced by an input. From this perspective, I hypothesized that easy inputs would induce relatively small internal state changes and accumulate shorter internal times, whereas difficult or error-prone inputs would form more complex hidden-state trajectories and accumulate longer internal times. In practice, the internal time measured by RiemannMamba showed a strong relationship with input difficulty and was larger for misclassified and corrupted inputs. These findings demonstrate that even when all inputs pass through the same computational depth, the amount of change they induce within the model may differ.

This research also showed that assessing model reliability solely through final-output accuracy or confidence is insufficient and that the internal-state changes formed while processing an input can serve as an important reliability signal. RiemannMamba’s internal time τ represents not the number of computational steps, but the accumulated geometric changes induced by each input in Mamba’s hidden-state space. Inputs that were difficult to classify, misclassified, or affected by distribution shift accumulated longer internal times, allowing the likelihood of model error to be evaluated from a perspective distinct from final confidence. In addition, Tau-CAM, which visualizes the spatial distribution of internal time, made it possible to observe which input regions induced large changes in the model’s internal state.

1.3 Driver Handover Readiness Estimation via Bio–Vehicle Signal Fusion and Safety–Oriented Re-decision

Safe Decision-Making in Human–Machine Collaboration

In conditional automated driving, transferring control of the vehicle back to the driver is not merely a driver-state classification problem, but a decision-making problem that must safely coordinate collaboration between a human and an automated system. Before handing control to the driver, the vehicle must determine whether the driver has recognized the current driving situation and is actually ready to resume driving. Transferring control too early to an unprepared driver increases the risk of an accident, while excessively delaying a necessary handover may also compromise safety.

I conducted a multimodal learning study that estimated driver handover readiness by jointly using the driver’s physiological signals and vehicle signals. Physiological signals reflect the driver’s cognitive and physical arousal and readiness, whereas vehicle signals describe the current dynamic state of the vehicle, including speed, steering,

acceleration, and deceleration. Because the two modalities provide information about different aspects of the situation, neither alone is sufficient to determine whether a safe handover is possible. I therefore integrated the two signal types to evaluate the states of the driver and vehicle together and to achieve more stable readiness estimation than would be possible with a single modality. However, high classification accuracy alone cannot guarantee safety in real driving environments. The driver's condition and the vehicle's driving state continue to change, meaning that a handover decision that was appropriate at the initial decision point may later become unsafe. I therefore studied a safety-oriented re-decision structure that does not fix the initial readiness estimate as the final decision, but instead reassesses handover feasibility using newly observed physiological and vehicle signals. When the model's prediction is uncertain or the driver's readiness is judged insufficient, the system withholds or reevaluates the transfer of control, preventing a single erroneous prediction from immediately leading to a hazardous action. This research taught me that the role of AI in systems where humans and machines collaborate is not simply to replace the human, but to coordinate when and how actions should be taken while jointly considering the human's condition and the system's limitations. In safety-critical environments, the ability to recognize uncertainty and revise an earlier decision in response to a changed state is more important than trusting a single prediction. This experience led me to view trustworthy AI not simply as a high-accuracy model, but as a system that can collaborate with people while preserving human control and safety.

1.4 Integration of My Research Experience

TST-Mamba investigated what information a model selects and preserves within spatiotemporal data, whereas RiemannMamba analyzed how strongly the selected information changes the model's internal state and how those changes relate to decision difficulty and reliability. Through these two studies, I came to believe that model failure does not suddenly emerge at the moment when the final output becomes incorrect. Rather, it can develop gradually through earlier processes of information selection and internal-representation change. The Driver Handover project further highlighted that when information selection and reliability estimation lead to an actual action, a system should not rely on a single prediction. Because the driver's condition and the vehicle's driving state continue to change, an initial handover decision cannot be assumed to remain safe. The system must therefore reevaluate its earlier decision using newly observed physiological and vehicle signals and withhold or revise it when it is no longer considered safe. Through this work, I came to view trustworthy AI not merely as a model with high accuracy, but as a system capable of observing changes in its internal state and predictive uncertainty and revising its decisions in response to changes in people and the environment.

Building on these research experiences, I aim to detect early changes that arise when a model begins to depart from its original task or the user's intention by analyzing the geometric trajectory of its internal states, and to control the corresponding representations and behavior so that they return to an appropriate direction.

2 Future Research

Understanding the Internal World of AI through Physics-Based White-Box Models

The most fundamental question I want to investigate is: What actually happens inside an AI model? Current deep learning models achieve high performance, yet it remains difficult for humans to determine why a model regarded particular information as important, what internal process led it to a conclusion, and at what point it began to form an incorrect judgment. Observing only the final output and confidence cannot sufficiently explain how that judgment was formed, and this opacity is the fundamental reason why contemporary AI models are still regarded as black boxes.

My long-term research goal is not merely to explain a model's final result after the fact, but to develop white-box AI in which the entire process can be observed, from the moment an input enters the model through the formation and transformation of internal representations to the final decision. I aim to view model reasoning not as an instantaneous output, but as a continuous process of state change occurring in hidden-state space. By analyzing which information enters the internal state, which information is suppressed, how different pieces of information are combined, and in which direction the representation moves as a result, I believe we can explain more fundamentally why a model arrived at a particular conclusion.

2.1 Physics-Based Interpretation of Internal AI Dynamics

The natural world is described by different physical theories, including classical mechanics, quantum mechanics, and relativity, depending on the conditions and scale involved. These theories do more than predict the outcomes of phenomena; they explain how the state of an object changes according to particular laws and how its current state gives rise to its next state. I believe that this physical perspective can also provide important insights into the internal representations of AI models.

The world directly perceived by an AI model is not the image or language itself as perceived by humans, but a hidden representation expressed as vectors and matrices. As an input passes through successive layers, it is represented as a point or state in a high-dimensional space and transformed into new states through the model's operations. This process can be viewed as formally analogous to the evolution of a physical system over time. I do not assume that the laws of nature can be applied directly and literally to neural networks. However, I believe that concepts such as state, trajectory, distance, energy, force, time, and conservation can be used to formalize internal model changes and experimentally test whether analogous structures exist. In this process, physics will serve not as a metaphor alone, but as a starting point for designing new measurable variables and model structures. My goal is to establish a theoretical and experimental framework that explains not only what a model outputs, but why a particular judgment was formed through the laws governing changes in its internal states.

2.2 White-Box AI that Reveals the Reasoning Process, Not Only the Result

Existing explainable AI methods largely focus on showing how strongly a particular image region or input token contributed to the final output. Although these methods can reveal what information the model referenced, they do not sufficiently explain how that information was processed internally, combined with other information, and transformed into the final reasoning process.

The white-box AI I aim to develop is not a model that merely provides an importance map. A user should be able to inspect the model's internal reasoning process step by step. For example, it should be possible to track what information the model accepted first, what information it quickly forgot, and how long a particular piece of information continued to influence later state changes. It should also be possible to observe how the model revised its initial hypothesis in response to new information and at which stage it formed its final judgment.

To achieve this, I plan to track hidden-state trajectories at the token, layer, and whole-input levels and transform internal changes into variables that humans can interpret. I will analyze model reasoning using geometric and dynamical indicators such as internal time, path length, changes in direction, curvature between states, stability, and convergence. Beyond identifying correlations, I also aim to conduct representation intervention and steering to test whether changes in particular internal states causally influence the final judgment. If modifying an internal representation changes the model's reasoning and output in a predictable direction, the role of that representation in the model's decision can be explained more clearly. Through this approach, I aim not only to observe a model's internal structure, but also to develop a white-box framework in which the concepts and reasoning paths learned by the model can be experimentally verified.

2.3 Safe and Trustworthy AI through Understanding Internal Processes

My desire to understand a model's internal processes is not motivated by academic curiosity alone. Even a highly accurate model cannot be fully trusted in society if its internal process cannot be inspected. Without such inspection, it is difficult to distinguish whether the model produced a correct answer by chance, reached a conclusion from incorrect grounds, or had its judgment manipulated by malicious external information.

I view the current development of AI as comparable to a new industrial revolution. The development of steam engines and mechanization profoundly transformed productivity and ways of life, while also causing changes in labor conditions, job displacement, safety accidents, and social conflict. Industry was able to continue developing not only because technological performance improved, but also because institutions and technologies for protecting workers and users were developed alongside it. AI is likewise unlikely to disappear from society; instead, it will become increasingly embedded throughout daily life and industry. We must therefore improve not only model capabilities, but also the safety and reliability required for people to use AI with confidence.

My concerns about these issues became clearer while studying law in preparation for the Korean patent attorney examination and while encountering a wide range of legal cases and legal and societal questions related to AI in a law school course. Technology itself may be neither inherently good nor evil, but opaque and difficult-to-control AI may be exploited for malicious purposes or make decisions that infringe upon human rights. As models become connected to personal information, external tools, and decision-making authority, an incorrect judgment may lead not merely to an erroneous answer, but to tangible harm. I therefore aim to develop methods for detecting the formation of dangerous reasoning inside a model at an early stage and correcting state changes that move in an unsafe direction. Model failure may not arise suddenly when the final output is generated, but may instead develop gradually through earlier processes of information selection and representation change. By observing trajectories of internal representations, it may be possible to identify when a model begins to deviate from its original task or the user's intention before that deviation is translated into action. After detection, I aim to study control methods that steer the model's state in a safer direction or halt the action and return the decision to a human when uncertainty is high. Ultimately, my goal is not merely to filter harmful outputs, but to develop AI that can examine its own internal reasoning process and correct a dangerous judgment before it leads to action.

2.4 Long-Term Research Goal

My research aims to understand the internal world of AI from a physical perspective and reveal it in a form that humans can observe and verify. Just as physics explains the principles of change underlying observable phenomena in nature, I seek to identify the laws governing changes in the internal representations underlying AI outputs. To this end, I plan to combine state-space modeling, Riemannian geometry, dynamical systems, and causal representation analysis to investigate how a model's hidden states are formed and transformed. I also aim to move beyond simply visualizing internal-state changes by formalizing them as interpretable variables connected to model reasoning, reliability, error likelihood, and safety.

Ultimately, I want to develop AI that does more than provide an answer. Humans should be able to determine what information the model used, what changes occurred internally, and why it reached a particular conclusion. The model should also be able to reveal when incorrect reasoning begins and stop or revise itself before that reasoning results in dangerous behavior. Drawing on ideas from physics, I aim to uncover the operating principles of the hidden-state world perceived by AI and, on this basis, develop safe white-box AI that is interpretable, controllable, and capable of exerting a positive influence on humanity.